

On my wavelength...

Graphic images of impending storms, instant reporting of events from remote corners of the world, receiving quality radio broadcast even when you are on the move...welcome to the world of satellites

Isn't it strange that satellites that were developed for spying and other defence needs during the Cold War are now affecting our daily lives? So much so that we don't leave our house for a trip to the nearest hill station before seeing the latest weather report beamed down to us by our 'friend from up above'—the INSAT IIB satellite.

When disaster strikes

Jan 2001: Devastating earthquake rocks Gujarat

Predicting earthquakes is an extremely complex task due to the challenges involved in tracking the minute shifting of the earth's tectonic plates, which collide to cause earthquakes. These measurements are difficult to record because the plates move just a few millimetres every year.

Seismologists at the California Institute of Technology and the Japanese Geographical Survey Institute (GSI) are now hoping to use Global Positioning Systems (GPS) to accurately predict an earthquake a few hours before it actually strikes, thus preventing the loss of thousands of lives.

Solar-powered receivers and antennas are placed in quake-prone regions. They transmit their positions through signals, which are sent out every 30 seconds or so, to two or more satellites in orbit. Researchers and scientists use the results to try and predict when the stress building between plates is likely to culminate into an earthquake in a particular region.

GPS locks into air traffic

November 1996: Midair collision over India. Over 300 feared dead

The exponential growth in the sheer volume of air traffic over the last decade has led to a number of new challenges for the existing air traffic control infrastructure. Managing the progress of each individual aircraft is currently done by tracking them using a combination of high-frequency radio and radar networks.

Boeing has put forward an audacious plan that proposes to replace radar with GPS-based satellite tracking. GPS receivers can pinpoint their location with a great degree of precision by triangulating signals from any of the 24 NAVSTAR satellites that orbit the earth. This really precise information can then theoretically be used to reduce the separation between aircraft and plan more direct routes to a destination.

Plans include using a constellation of 16 satellites in a 12-hour medium earth orbit (MEO) at the same altitude as the NAVSTAR GPS satellite system. Such a large network would help ensure that any airplane or air traffic control terminal is within the view of at least two satellites, so that in the event of a satellite failure, an adjacent satellite can ensure seamless service.

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